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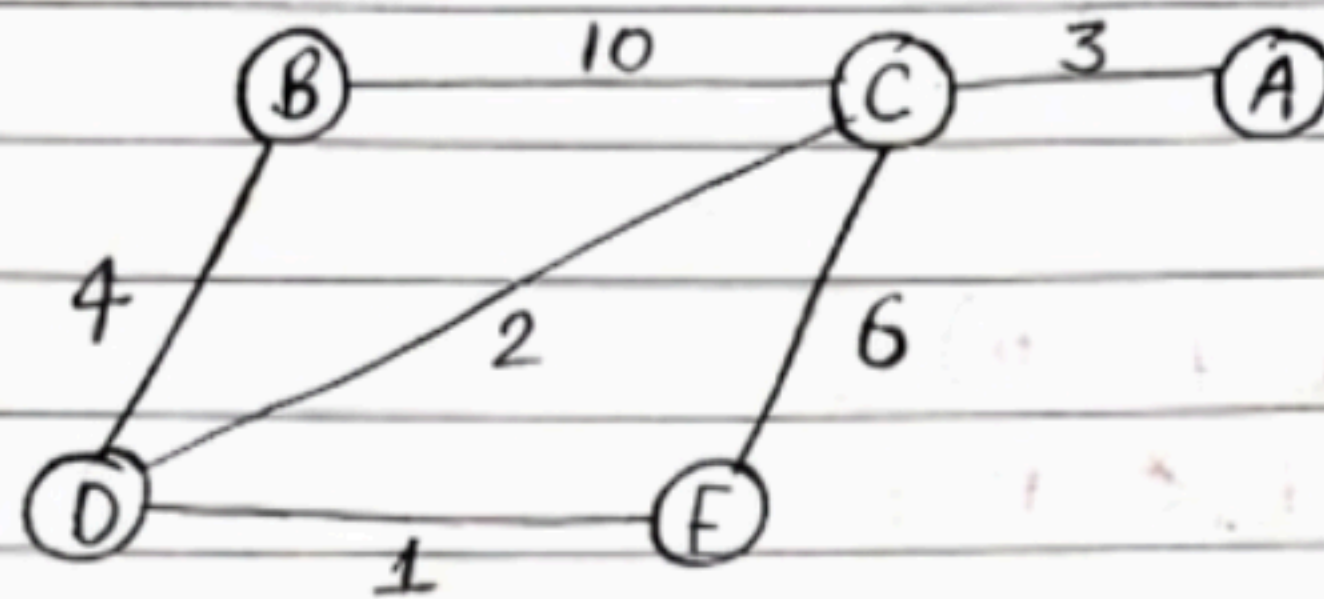
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Assignment - 03

Q1. Consider the following Graph G and obtain the minimum Spanning tree using Prim's Approach?



Sol:

$$B-C = 10$$

$$C-A = 3$$

$$C-D = 4$$

$$D-C = 2$$

$$C-E = 6$$

$$D-E = 1$$

Start from any node

Steps 1: Start from A

- Select A-C

$$MST = \{A, C\}$$

$$Cost = 3$$

Step 2: From $\{A, C\}$

- $C-B = 10$

- $C-D = 2$

- $C-E = 6$

Minimum = $C-D (2)$

$$*MST = \{A, C, D\}$$

$$Cost = 3 + 2 = 5$$

Step 3: From $\{A, C, D\}$

- $D-E = 1$

- $D-B = 4$

- $C-B = 10$

- $C-E = 6$



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$$\text{Minimum} = D-E (1)$$

$$\text{MST} = \{A, C, D, E\}$$

$$\text{Cost} = 5 + 1 = 6$$

Step 4 : From $\{A, C, D, E\}$

$$\bullet D-B = 4$$

$$\bullet C-B = 10$$

$$\text{Minimum} = D-B (4)$$

$$\text{MST} = \{A, B, C, D, E\}$$

$$\text{Cost} = 6 + 4 = 10$$

Final MST

$$\bullet A-C = 3$$

$$\bullet C-D = 2$$

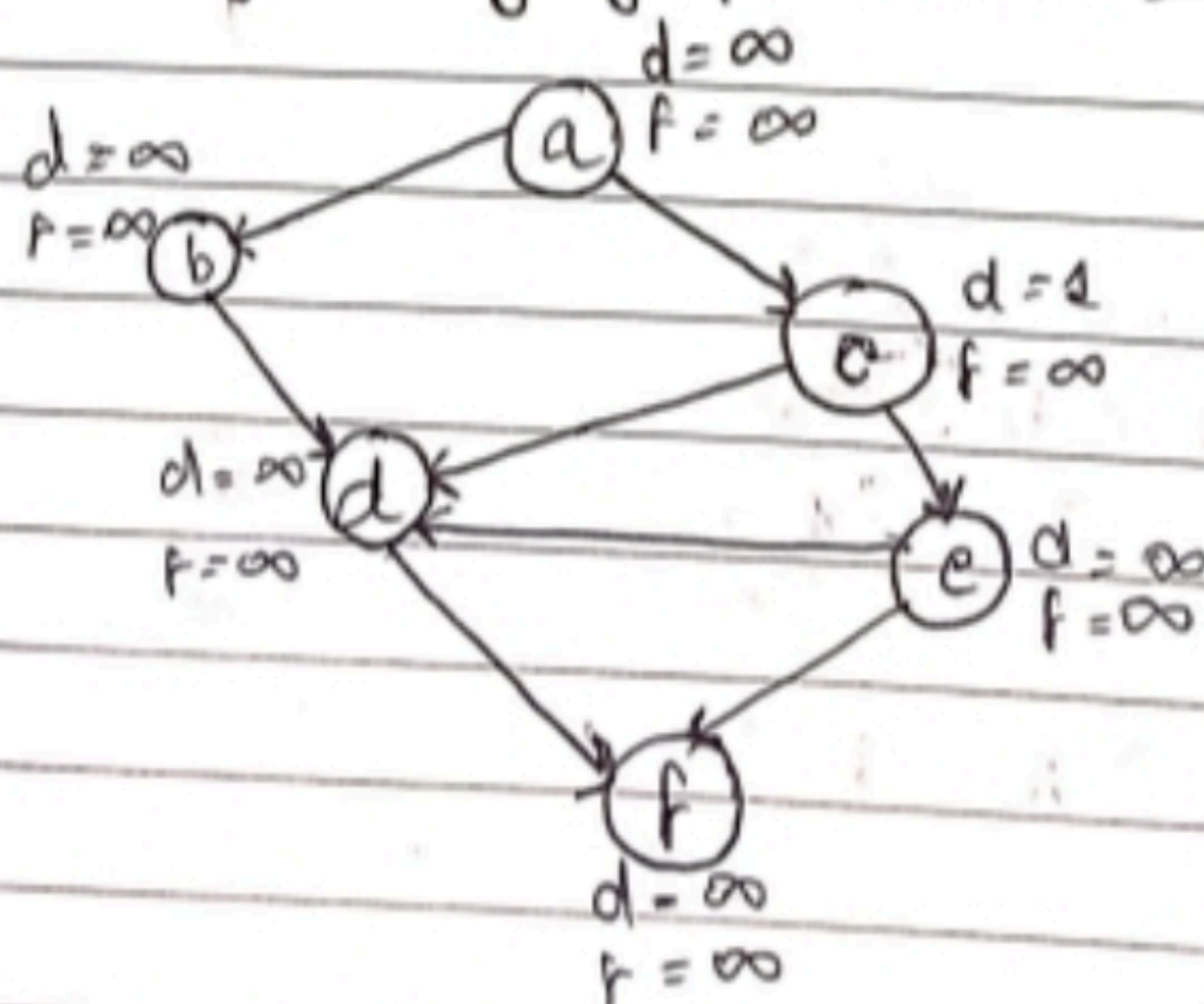
$$\bullet D-E = 1$$

$$\bullet D-B = 4$$

Total Minimum Cost = 10

Q2

Consider the following graph G_1 . Compute the topological sort of the following graph where starting vertex is C ?



Sol: Step 1: Start DFS from C

Visit order:

- $C \rightarrow D \rightarrow F$

Push to Stack:

- F, D

Step 2: Back to C $\rightarrow$ Visit E

- E - F (already visited)

Push:

- E

Then push:

- C

Step 3: Visit remaining nodes (A, B)

Visit A:

- $A \rightarrow B \rightarrow D$ (visited)

Push:

- B

- A

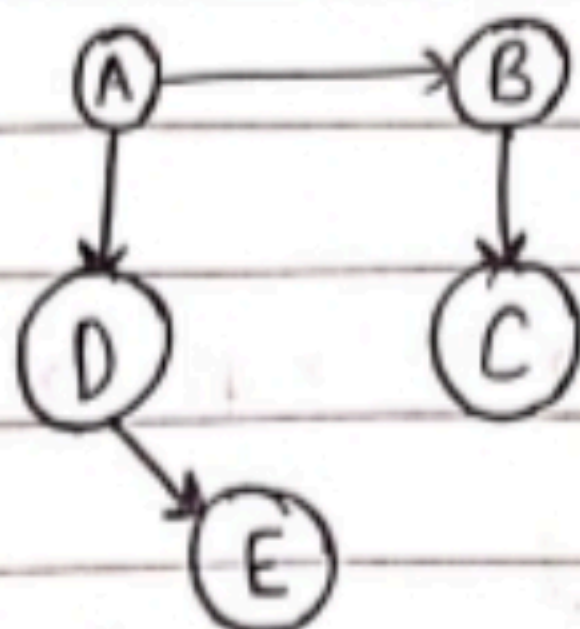
Stack (Topological Order):

Reverse of Stack: $A \rightarrow B \rightarrow C \rightarrow E \rightarrow D \rightarrow F$

Topological Sort = A B C E D F

Q3

What would be the DFS and BFS traversal of the given graph?



Solⁿ:- (1) BFS Traversal (Level wise)
Start from A

Step by step :

Step 1 :

- Start: A
- Visit neighbour of A $\rightarrow$ B, D

Step 2 :

- Visit neighbour of B
- B $\rightarrow$ C

Step 3 :

- Visit neighbour of D
- D $\rightarrow$ E

BFS Order : A $\rightarrow$ B $\rightarrow$ D $\rightarrow$ C $\rightarrow$ E

(2) DFS Traversal (Depth wise)

Start from A

- &

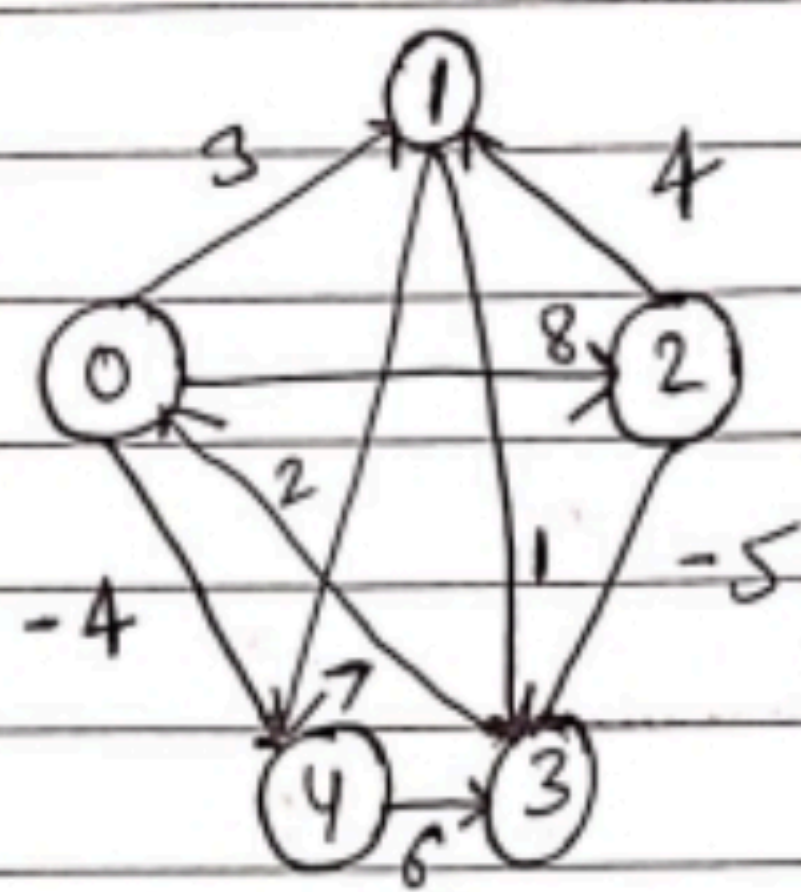
Step 1 :

- Start : A

- ① Visit A
- ② Go to B
- ③ Go to C
- ④ Backtrack to A
- ⑤ Go to D
- ⑥ Go to E

DFS Traversal Order : A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D $\rightarrow$ E

Q4 Consider the graph G apply the Floyd Warshall algorithm on the graph



	1	2	3	4	5
1	0	3	8	∞	-4
2	∞	0	∞	1	7
3	∞	4	0	∞	∞
4	2	∞	-5	0	∞
5	∞	∞	∞	6	0

Solⁿ Given weight Matrix W :

$$W = \begin{bmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & \infty & -5 & 0 & \infty \\ \infty & \infty & \infty & 0 & 0 \end{bmatrix}$$

Floyd Warshall Formula:

$$D_{ij}^{(k)} = \min(D_{ij}^{(k-1)}, D_{ik}^{(k-1)} + D_{kj}^{(k-1)})$$

Update matrix by considering each node as intermediate ($k=1$ to 6)

After $k=1$ (Node 1 as intermediate)

Update paths using node 1

After $k=2$ (Node 2):

- update shortest paths via node 2

After $k=3$ (Node 3):

- update using node 3.



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After $K=4$ (Node 4) :

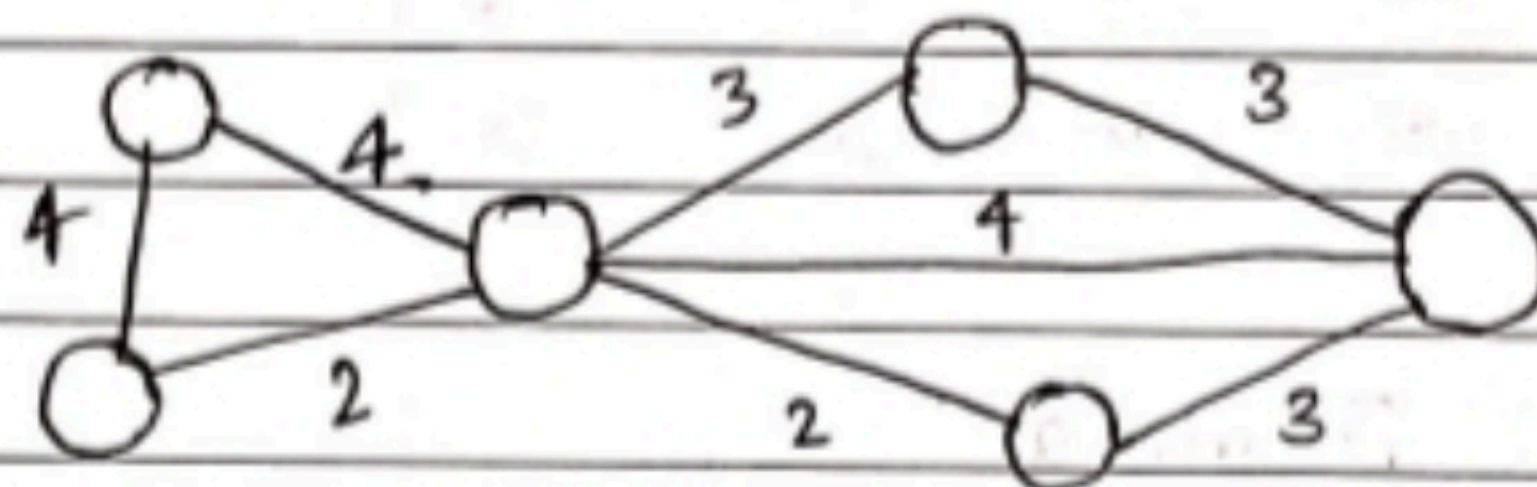
updates (eg. negative weights improve paths)

After $K=5$ (Node 5) :

• Final Refinement

$$D = \begin{bmatrix} 0 & 1 & -3 & 2 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 7 & 4 & 0 & 5 & 3 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{bmatrix}$$

Q5 Consider the following Graph G and obtain the minimum spanning tree using Kruskal Approach



Sol:-

Step 1 : Identify Edges with weights

Edges:

• (Left bottom - centre) = 2

(centre - bottom right) = 2

(C - t.m) = 3

(t.m - r) = 3

(B.r - r) = 3

(t.l - c) = 4

(Left vertical) = 4

(C - r) = 4

Step 2: Sort Edges

2, 2, 3, 3, 3, 4, 4, 4

Step 3: Apply Krushkal

pick smallest edges one by one:

1. Add edge (2)

2. Add edge (2)

3. Add edge (3)

4. Add edge (3)

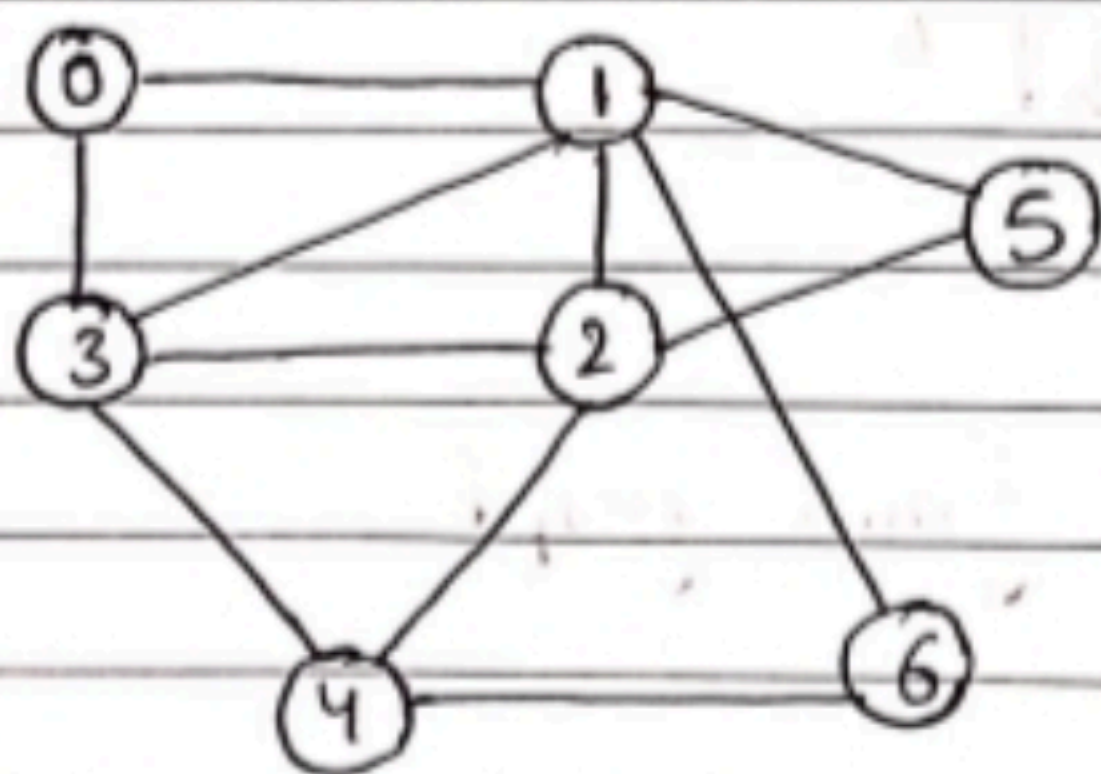
5. Add edge (3) → Now all vertices connected
(Stop when $n-1$ edges = $6-1 = 5$ edges)

MST Edges Selected : 2, 2, 3, 3, 3

Total Cost = $2 + 2 + 3 + 3 + 3 = 13$

MST Cost = 13

Q6 Compare BFS and DFS algo on the basis of time and space complexity using the graph given below:



Time Complexity :

Algorithm

BFS

DFS

Time Complexity

$O(V+E)$

$O(V+E)$

→ Both visit all vertices & edges once.

Algorithm

BFS

DFS

Space Complexity

$O(V)$ (Queue)

$O(V)$ (Stack/Recursion)

Basis

Traversal

Data Structure

Shortest path

Space

BFS

Level-Basis

Queue

Yes (unweighted)

More (wide graph)

DFS

Depth-wise

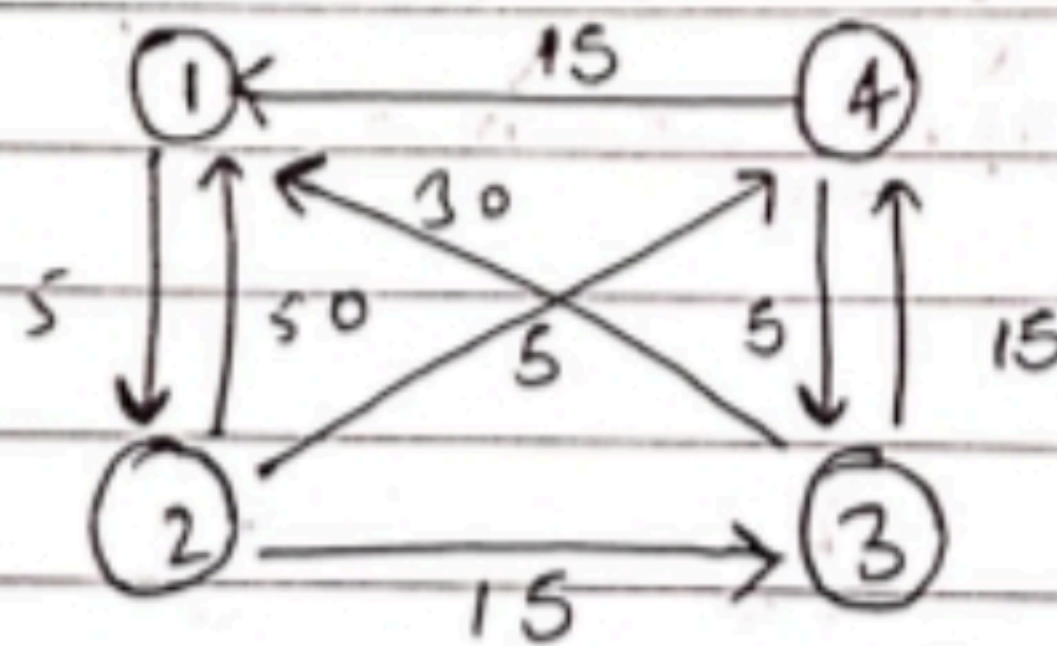
Stack

No

Less

Q7

Consider the graph G apply the Floyd Warshall algo on the



Solⁿ →

Step 1 : Identify edges from graph

• $1 \rightarrow 2 = 5$

$2 \rightarrow 1 = 50$

$2 \rightarrow 3 = 15$

$3 \rightarrow 4 = 5$

$4 \rightarrow 3 = 15$

$4 \rightarrow 1 = 15$

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$$1 \rightarrow 3 = 30$$

$$2 \rightarrow 4 = 5$$

Initial Matrix

$$D = \begin{bmatrix} 0 & 5 & 30 & \infty \\ 50 & 0 & 15 & 5 \\ \infty & \infty & 0 & 15 \\ 15 & \infty & 5 & 0 \end{bmatrix}$$

$$D[i][j] = \min(D[i][j], D[i][k] + D[k][j])$$

$$D = \begin{bmatrix} 0 & 5 & 20 & 10 \\ 20 & 0 & 10 & 5 \\ 30 & 35 & 0 & 15 \\ 15 & 20 & 5 & 0 \end{bmatrix}$$